

# Tide retrospective: the identifiability lower bound

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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## 1 Setting

Closing tide of the three-tide germbij arc, after 2026-08-08-19-11-tide-germbij-pencil (the pencil identity) and 2026-08-09-00-59-tide-germbij-sector (the sector bound). The germbij note proves that for nonnegative real analytic losses with common zero locus, equality of all Boltzmann-integral expansions forces equality of germs; the engine is a single quantitative inequality, and this tide lands it in Lean in one dimension. With this, the mathematical content of the note's Theorem 7.3 chain (in the 1D, vanishing-at-a-point setting, with the analytic input factored into a finite-vanishing-order hypothesis) is formalised end to end.

## 2 The thing formalised

One theorem, `pencil_difference_lower_bound` in `Laplace/Identifiability.lean`, sorry-free. For  $L_1, L_2 \geq 0$  with  $L_1 + L_2 \leq C_0 w^2$  on  $[0, r_0]$ ,  $g = L_2 - L_1$  with  $c w^m \leq |g|$  on  $[0, r_0]$ , and  $\psi \geq 0$  with  $\psi \equiv 1$  on  $[0, r_0]$ , under  $4 \leq r_0^2 t$  and explicit integrability hypotheses (the minorant, the uncurried pencil integrand on  $[0, 1] \times \mathbb{R}$ , and each pencil slice):

$$c^2 e^{-4C_0} \cdot t \cdot t^{-m-1/2} \leq \int_{\mathbb{R}} g(w) \psi(w) (e^{-tL_1(w)} - e^{-tL_2(w)}) dw.$$

Since the right-hand side would be  $o(t^{-N})$  for every  $N$  if the two Boltzmann families had equal expansions, and the left-hand side is a fixed positive power of  $t$ , this is the contradiction proving 1D identifiability: distinct potentials of this class are distinguished by the single observable  $g\psi$ . The numerical check (tide log):  $L_1 = w^2/2$ ,  $L_2 = w^2/2 + w^4$ ,  $t = 100$  gives  $\Delta = 1.8 \times 10^{-5}$  against the bound  $3.4 \times 10^{-11}$ .

## 3 Proof strategy

Six steps, each one landed lemma or one Mathlib call. The pencil identity (previous tide) rewrites the difference as

$$t \int_0^1 \int g^2 \psi e^{-tL_s} dw ds,$$

and each slice dominates the  $s$ -independent minorant  $\int g^2 \psi e^{-t(L_1+L_2)}$ , by the pencil comparison lemma and integral monotonicity. The slice function is interval integrable because the Fubini hypothesis slices<sup>1</sup>, so integrating the constant bound over  $s \in [0, 1]$  preserves it.<sup>2</sup> The minorant

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<sup>1</sup>`Integrable.integral_prod_left` followed by `intervalIntegrable_iff` and `Set.uIoc_of_le`.

<sup>2</sup>`intervalIntegral.integral_mono_on`.

dominates its restriction to the Laplace-scale window, the integrand being nonnegative;<sup>3</sup> on the window  $\psi = 1$  (the window sits inside  $[0, r_0]$  exactly when  $4 \leq r_0^2 t$ ), and the sector bound of the previous tide finishes with  $K = L_1 + L_2$ ,  $a = g$ .

## 4 Roadblocks and resolutions

One compile failure in total: a `rw` inside the `setIntegral_congr_fun` step could not see the pattern  $\psi w$  through the unreduced lambda application that `EqOn` introduces; `simp only` with the same rewrite, which beta-reduces first, fixed it. The stacked-branch mechanics (this tide chains off the sector tide’s branch, which merged mid-tide) resolved with one clean rebase onto the advanced `main`.

## 5 What was Mathlib, what was new

Mathlib supplied all the monotonicity and slicing infrastructure (integral monotonicity in its plain, set-restricted, and interval forms, and the product-measure slicing lemma). New content: the composition itself, and the hypothesis discipline: every integrability assumption is explicit, and the Fubini hypothesis is reused to derive interval integrability of the slice function rather than assuming it separately.

## 6 Lessons

Two additions to the playbook. First, `rw` under `EqOn` goals: use `simp only` for rewrites after `setIntegral_congr_fun`, since the pointwise goal arrives as an unreduced application. Second, a product-integrability hypothesis is worth more than it looks: it feeds both the Fubini exchange and, via slicing, the one-dimensional integrability needed by interval monotonicity, so one hypothesis serves two proof obligations.

## 7 Follow-ups

- Weaken the per-slice integrability hypothesis to its almost-everywhere consequence of the product hypothesis (the interval monotonicity step then needs its a.e. variant).
- An instantiation lemma: derive all three integrability hypotheses from continuity of  $L_1, L_2$  plus compact support of  $\psi$ , packaging the theorem for direct application to concrete potentials.
- The asymptotic packaging: state the contradiction itself, e.g. the difference is not  $o(t^{-N})$  for  $N$  past the exponent, as an `Asymptotics.IsLittleO` negation, connecting to the seabed’s asymptotics idiom.
- Beyond 1D: the multivariate sector bound over a spherical-cap sector, toward the full germbij Theorem 7.3.

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<sup>3</sup>`MeasureTheory.setIntegral.le_integral`.